

## CEN3024C Module 8 Project

For this project we will enhance the TaskManager application by implementing automated testing and continuous integration (CI) using GitHub Actions. Building on previous work, we will create and verify unit tests, then automate the build and test process through a workflow file in the repository.

### 1. Create Task Tests:

- Use the UserRepository and UserService test examples in the GitHub Classroom repository as references.
- Implement new tests for the TaskRepository and TaskService classes.
- For TaskRepository: create both an in-memory test class and a mocking test class.
- For TaskService: create one mocking test class.

### 2. Verify Tests

- Right-click on the java subfolder under the test directory in IntelliJ and select Run all tests.
- Ensure all tests display green checkmarks indicating success.
- Take a snip of the test results for submission

### 3. Create a GitHub Actions Workflow

- Under your repository's .github/workflows folder, create a new file named build-and-test.yml.
- The workflow should automatically build and test the application when a .java file is added or modified.

#### Sample Workflow Content

```
name: Build and Run TaskManager Application

on:
  push:
    paths:
      - '**/*.java'
  pull_request:
    paths:
      - '**/*.java'
  workflow_dispatch: # Enables manual trigger

jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3

      - name: Set up JDK 17
        uses: actions/setup-java@v3
        with:
          java-version: '17'
          distribution: 'temurin'

      - name: Run Maven Tests
        run: mvn -f pom.xml test
```

```

build-and-run:
  needs: test
  runs-on: ubuntu-latest
  steps:
    - uses: actions/checkout@v3

    - name: Set up JDK 17
      uses: actions/setup-java@v3
      with:
        java-version: '17'
        distribution: 'temurin'

    - name: Build with Maven
      run: mvn -f pom.xml clean package -DskipTests

    - name: Run Spring Boot Application
      run: |
        java -jar target/*.jar &
        echo $! > springboot.pid
      timeout-minutes: 5

    - name: Shutdown Spring Boot Application
      run: kill $(cat springboot.pid)

```

#### 4. Workflow Verification

- Commit and push your .yml workflow file.
- Push a change (this can be as simple as adding or changing a comment in a .java file).
- Navigate to the **Actions** tab in GitHub to verify the automated build and test process.
- Demonstrate a successful test run (either automatic or manual) using a screenshot of the executed workflow

#### Submission

Submit your project and screensnips showing your test results to the GitHub classroom repo.